

Konstantinos Garyfallogiannis

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Education

- **Ph.D. in Mechanical Engineering**, University of Pennsylvania 2019 - 2023
- **Master of Science in Robotics**, University of Pennsylvania 2020 - 2022
- **Master of Science in Mechanical Engineering**, University of Thessaly 2017 - 2019
- **Bachelor of Science in Mechanical Engineering**, University of Thessaly 2014 - 2017

Experience

Research Fellow at University of Pennsylvania, School of Engineering 2019 - September 2023

- Collaborated with an 8-people multi-disciplinary team in a \$10 million research project
- Translated problems to models for analysis based on observation, and theory.
- Performed static and **dynamic** (modal) mechanical, thermal, and thermo-mechanical analysis for non-linear problems using Finite Elements
- Performed fatigue, fracture and failure analysis using Finite Elements
- Devised a methodology for studying dynamic behavior of meta-materials
- Mentored new lab members and delegated projects across the younger members of the lab
- Presented work in conferences, seminars, and workshops and interacted with other participants

Robotics Graduate Student at University of Pennsylvania, School of Engineering 2020 - 2022

- Programmed the kinematics, and control of a robotic arm, and won **2nd** place to the university competition
- Analyzed dynamics of a quadcopter, *developed the control codes*, and won **2nd** place in course competition

Student Researcher at University of Thessaly, School of Engineering 2017 - 2019

- Conceptualized and published a finite element framework to predict composite material properties and showed <5% error with commercially available data.

Intern structural engineer at Sielman S.A. June – September 2018

- Aided the feasibility studies, design, optimization and specification of materials and procedures needed for mechanical systems, mainly vessel parts and mechanisms of rockets and airplanes.

Engineering Softwares

SolidWorks, Abaqus, Ansys, AutoCAD, Autodesk Inventor, Python, Matlab, Mathematica, Microsoft Office

Teaching experience

Teacher assistant in "Feedback Control", University of Pennsylvania 2021

- Taught classic controls, like poles' dynamics, Bode diagrams, Nyquist plots, PID controllers to more advanced concepts, like spectral decomposition, state-space representation, and of nonlinear systems

Teacher assistant in "Mechanics of Solids", University of Pennsylvania 2020 & 2021

- Instructed students to simplify and adapt engineering problems to simple concepts taught in the class. Facilitated sessions to provide students with hands-on design and finite element experience

Leadership Experience

Vice President of Graduate student association at University of Pennsylvania 2021-2022

- Organized 7 informative and social events and aided 6 students with academic and personal problems.

President of Undergraduate student association in University of Thessaly 2015-2016

- Organized 4 informative events and mediated for 2 students on the faculty board.

Scholarships and awards

- *John A. Goff award*, University of Pennsylvania 2023
Awarded on the criteria of resourcefulness, scholarship, and leadership (2500\$).
- *Gerondelis Foundation Scholarship*, University of Pennsylvania 2020
Awarded to Greek citizens pursuing pioneering research abroad (6000\$).